4 **(a)** Define electric potential at a point.

.....

.....

..... [2]

(b) A hydrogen atom may be considered to consist of a proton and an electron separated by a distance of 120 pm, as shown in Fig. 4.1.

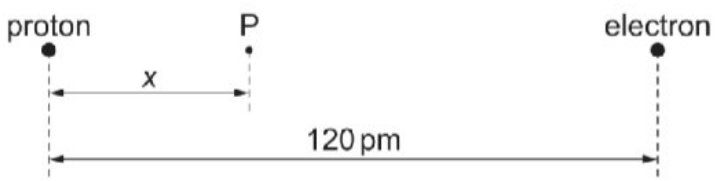


Fig. 4.1

The two particles may be considered as point charges.

Point P lies on the line joining the electron and the proton and is at a variable distance x from the proton.

(i) **1.** Show that the electric potential V at point P when x = 10 pm is equal to 130 V.

[2]

2. Calculate V when x = 30 pm.

V = ...V [1]

3. On Fig. 4.1, draw a cross (×) at one position, other than infinity, where the electric potential is zero. [1]

4. On Fig. 4.2, sketch the variation of V with x between x = 10 pm and x = 110 pm. [3]

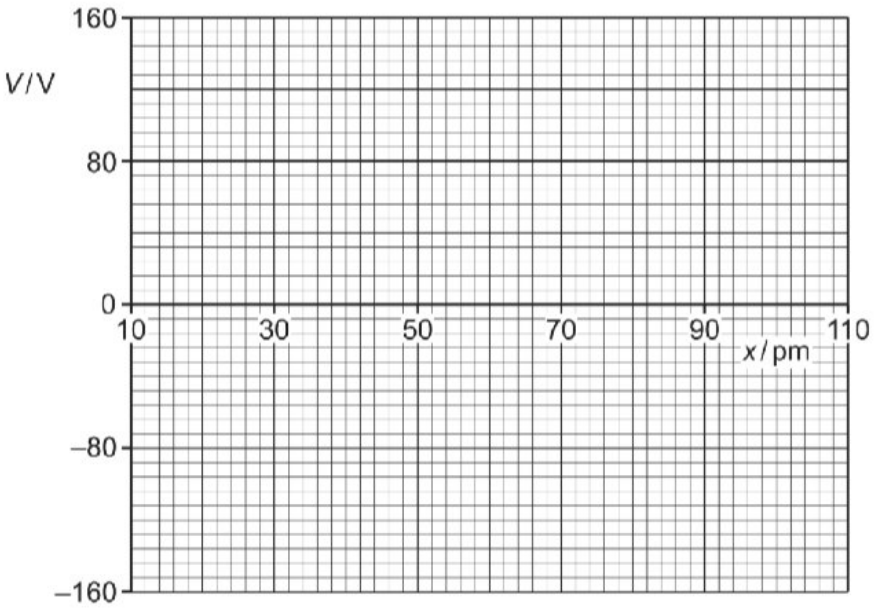


Fig. 4.2

(ii) Calculate the resultant electric field strength at the position identified in **(b)(i)3**.

electric field strength = ... N C^{-1} [2]

(iii) Fig. 4.3 shows the hydrogen atom placed in a non-uniform electric field E, for which field lines are shown. Describe the initial motion of the hydrogen atom at this instant.

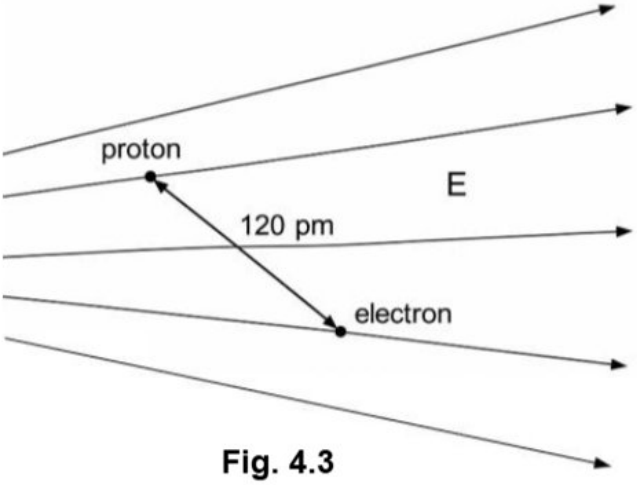


Fig. 4.3

.....

..... [2]